Mathematical Foundations of Political Methodology

Robert Gulotty

Associate Professor
Department of Political Science
University of Chicago

August 28th, 2017

- Determinant
- Eigenvalues and Eigenvectors
- Multivariate Function
- Multivariate Derivative

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Multivariable Calculus

Functions of many variables:

1) Policies may be multidimensional (policy provision and pork buy off)
2) Countries may invest in offensive and defensive resources for fighting wars
3) Ethnicity and resources could affect investment

Today:

1) Determinant
2) Eigenvector/Diagonalization
3) Multivariate functions
4) Partial Derivatives, Gradients, Jacobians, and Hessians

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Determinant

Suppose we have a (2×2) matrix A

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

The determinant of A for this 2×2 matrix, which is written as $|A|$ or $det(A)$, is defined as: $a_{11}a_{22} - a_{21}a_{22}$.

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Now, suppose we have a $n \times n$ matrix A

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Define ij-entry, written a_{ij}, as the number in the i-th row and j-th column.

Define ij-minor, written $|A_{ij}|$, is the determinant that's left after deleting from $|A|$ the row and column containing a_{ij}.

Define ij-cofactor, written as A_{ij}, is given as a formula by $A_{ij} = (-1)^{i+j}|A_{ij}|$.

Now, the $det(A)$ is defined as: $det(A)(or|A|) = \sum_{j=1}^{n} a_{ij}A_{ij} = \sum_{i=1}^{n} a_{ij}A_{ij}$

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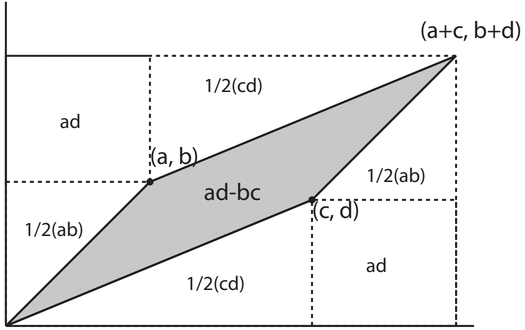
For 2×2 matrices they have a natural interpretation:



The area of the parallelogram is

$$|(a+c)(b+d) - 2ad - ab - cd| = |ab + ad + bc + cd - 2ad - ab - cd| = |ad - bc|$$

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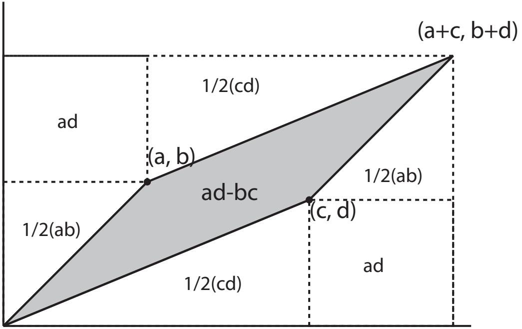

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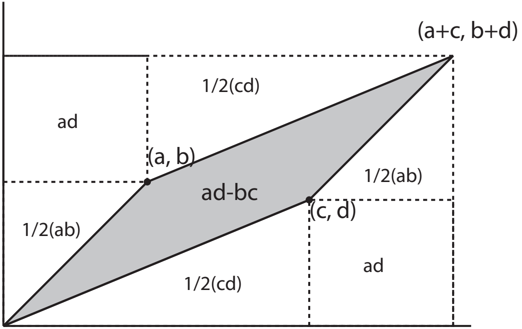

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$$A = \begin{pmatrix} 1 & 0 & 3 \\ 1 & 2 & -1 \\ 2 & 1 & -1 \end{pmatrix}$$

Then, the determinant by expanding the first row is:

$$|A| = 1 \times \begin{vmatrix} 2 & -1 \\ 1 & -1 \end{vmatrix} - 0 \times \begin{vmatrix} 1 & -1 \\ 2 & -1 \end{vmatrix} + 3 \times \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix}$$

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An Introduction to Eigenvectors, Values, and Diagonalization

Definition

Suppose $\boldsymbol{A}$ *is an* $N \times N$ *matrix and* λ *is a scalar.*
If

$$\boldsymbol{A}\boldsymbol{x} = \lambda \boldsymbol{x}$$

Then $\boldsymbol{x}$ *is an eigenvector and* λ *is the associated eigenvalue*

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An Introduction to Eigenvectors, Values and Diagonalization

After we define invertibility, eigenvalues and eigenvectors, a natural question would be: what is the relationship between those concepts.

Following theorem gives us the answer.

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Theorem

Suppose $\boldsymbol{A}$ is an invertible $N \times N$ matrix and further suppose that $\boldsymbol{A}$ has N distinct eigenvalues and N linearly independent eigenvectors. Then we can write $\boldsymbol{A}$ as,

$$\boldsymbol{A} = \boldsymbol{W} \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_N \end{pmatrix} \boldsymbol{W}^{-1}$$

where $\boldsymbol{W} = (\boldsymbol{w}_1, \boldsymbol{w}_2, \dots, \boldsymbol{w}_N)$ is an $N \times N$ matrix with the N eigenvectors as column vectors.

Proof:
Note

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\begin{aligned}
\boldsymbol{AW} &= \begin{pmatrix} \lambda_1 \boldsymbol{w}_1 & \lambda_2 \boldsymbol{w}_2 & \dots & \lambda_N \boldsymbol{w}_N \end{pmatrix} \\
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Examples of Diagonalization

Suppose $\boldsymbol{A}$ is an $N \times N$ invertible matrix with eigenvalues $\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \ldots, \lambda_N)$ and eigenvectors $\boldsymbol{W}$. Calculate $\boldsymbol{AA} = \boldsymbol{A}^2$

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$$f(x_1, x_2) = \sin(x_1)\cos(x_2)$$

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$$f(x_1, x_2) = -(x-5)^2 - (y-2)^2$$

Multivariate Functions

$$f(x_1, x_2, x_3) = x_1 + x_2 + x_3$$

Multivariate Functions

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\begin{aligned}
f(\boldsymbol{x}) &= f(x_1, x_2, \ldots, x_N) \\
&= x_1 + x_2 + \ldots + x_N \\
&= \sum_{i=1}^{N} x_i
\end{aligned}
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Multivariate Functions

Definition

Suppose $f : R^n \to R^1$. We will call f a multivariate function. We will commonly write,

$$f(\boldsymbol{x}) \quad = \quad f(x_1, x_2, x_3, \ldots, x_n)$$

- $R^n = R \underbrace{\times}_{\text{cartesian}} R \times R \times \ldots R$
- The function we consider will take n inputs and output a single number (that lives in R^1, or the real line)

Example 1

$$f(x_1, x_2, x_3) = x_1 + x_2 + x_3$$

Evaluate at $\boldsymbol{x} = (x_1, x_2, x_3) = (2, 3, 2)$

$$\begin{aligned} f(2,3,2) &= 2 + 3 + 2 \\ &= 7 \end{aligned}$$

Example 1

$$f(x_1, x_2) = x_1 + x_2 + x_1 x_2$$

Evaluate at $\boldsymbol{w} = (w_1, w_2) = (1, 2)$

$$\begin{aligned} f(w_1, w_2) &= w_1 + w_2 + w_1 w_2 \\ &= 1 + 2 + 1 \times 2 \\ &= 5 \end{aligned}$$

Preferences for Multidimensional Policy

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$$\begin{aligned} U(\boldsymbol{x}) &= U(x_1, x_2, \ldots, x_N) \\ &= -(x_1 - \mu_1)^2 - (x_2 - \mu_2)^2 - \ldots - (x_N - \mu_N)^2 \\ &= -\sum_{j=1}^{N}(x_j - \mu_j)^2 \end{aligned}$$

Suppose $\boldsymbol{\mu} = (\mu_1, \mu_2, \ldots, \mu_N) = (0, 0, \ldots, 0)$. Evaluate legislator's utility for a policy proposal of $\boldsymbol{m} = (1, 1, \ldots, 1)$.

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Preferences for Multidimensional Policy

Recall that in the spatial model, we suppose policy and political actors are located in a space.

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&= -\sum_{j=1}^{N} 1 = -N
\end{aligned}
$$

Multivariate Derivative

Definition

Suppose $f : X \to \Re^1$, *where* $X \subset \Re^n$. $f(\boldsymbol{x}) = f(x_1, x_2, \ldots, x_N)$. *If the limit,*

$$\begin{aligned} \frac{\partial}{\partial x_i} f(\boldsymbol{x}_0) &= \frac{\partial}{\partial x_i} f(x_{01}, x_{02}, \ldots, x_{0i}, x_{0i+1}, \ldots, x_{0N}) \\ &= \lim_{h \to 0} \frac{f(x_{01}, x_{02}, \ldots, x_{0i} + h, \ldots x_{0N}) - f(x_{01}, x_{02}, \ldots, x_{0i}, \ldots, x_{0N})}{h} \end{aligned}$$

exists then we call this the partial derivative of f *with respect to* x_i *at the value* $\boldsymbol{x}_0 = (x_{01}, x_{02}, \ldots, x_{0N})$.

Rules for Taking Partial Derivatives

Partial Derivative: $\frac{\partial f(\boldsymbol{x})}{\partial x_i}$

- Treat each instance of x_i as a variable that we would differentiate before
- Treat each instance of $\boldsymbol{x}_{-i} = (x_1, x_2, x_3, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n)$ as a constant

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Example Partial Derivatives

$$
\begin{aligned}
f(\boldsymbol{x}) &= f(x_1, x_2) \\
&= x_1 + x_2
\end{aligned}
$$

Partial derivative, with respect to x_1 at (x_{01}, x_{02})

$$
\begin{aligned}
\frac{\partial f(x_1, x_2)}{\partial x_1}|_{(x_{01}, x_{02})} &= 1 + 0|_{x_{01}, x_{02}} \\
&= 1
\end{aligned}
$$

Example Partial Derivatives

$$\begin{aligned} f(\boldsymbol{x}) &= f(x_1, x_2, x_3) \\ &= x_1^2 \log(x_1) + x_2 x_1 x_3 + x_3^2 \end{aligned}$$

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\end{aligned}
$$

What is the partial derivative with respect to x_1? Evaluated at $\boldsymbol{x}_0 = (x_{01}, x_{02}, x_{03})$.

$$
\begin{aligned}
\frac{\partial f(\boldsymbol{x})}{\partial x_1}|_{\boldsymbol{x}_0} &= 2x_1 \log(x_1) + x_1^2 \frac{1}{x_1} + x_2 x_3|_{\boldsymbol{x}_0} \\
&= 2x_{01} \log(x_{01}) + x_{01} + x_{02} x_{03}
\end{aligned}
$$

Example Partial Derivatives

$$\begin{aligned} f(\boldsymbol{x}) &= f(x_1, x_2, x_3) \\ &= x_1^2 \log(x_1) + x_2 x_1 x_3 + x_3^2 \end{aligned}$$

What is the partial derivative with respect to x_1? x_2? Evaluated at $\boldsymbol{x}_0 = (x_{01}, x_{02}, x_{03})$.

$$\begin{aligned} \frac{\partial f(\boldsymbol{x})}{\partial x_2}\Big|_{\boldsymbol{x}_0} &= x_1 x_3 \big|_{\boldsymbol{x}_0} \\ &= x_{01} x_{03} \end{aligned}$$

Example Partial Derivatives

$$\begin{aligned} f(\boldsymbol{x}) &= f(x_1, x_2, x_3) \\ &= x_1^2 \log(x_1) + x_2 x_1 x_3 + x_3^2 \end{aligned}$$

What is the partial derivative with respect to x_1? x_2? x_3? Evaluated at $\boldsymbol{x}_0 = (x_{01}, x_{02}, x_{03})$.

$$\begin{aligned} \frac{\partial f(\boldsymbol{x})}{\partial x_3}|_{\boldsymbol{x}_0} &= x_1 x_2 + 2x_3|_{\boldsymbol{x}_0} \\ &= x_{01} x_{02} + 2x_{03} \end{aligned}$$

Rate of Change, Linear Model

Suppose we regress Approval_i rate for Obama in month i on Employ_i and Gas_i. We obtain the following model:

$$\text{Approval}_i = 0.8 - 0.5\text{Employ}_i - 0.25\text{Gas}_i$$

We are modeling $\text{Approval}_i = f(\text{Employ}_i, \text{Gas}_i)$. What is partial derivative with respect to employment?

$$\frac{\partial f(\text{Employ}_i, \text{Gas}_i)}{\partial \text{Employ}_i} = -0.5$$

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Gradient

Definition

Suppose $f : X \to \Re^1$ with $X \subset \Re^n$ is a differentiable function. Define the gradient vector of f at $\boldsymbol{x}_0$, $\nabla f(\boldsymbol{x}_0)$ as,

$$\nabla f(\boldsymbol{x}_0) = \left(\frac{\partial f(\boldsymbol{x}_0)}{\partial x_1}, \frac{\partial f(\boldsymbol{x}_0)}{\partial x_2}, \frac{\partial f(\boldsymbol{x}_0)}{\partial x_3}, \ldots, \frac{\partial f(\boldsymbol{x}_0)}{\partial x_n}\right)$$

- The gradient points in the direction that the function is increasing in the fastest direction
- We'll use this to do optimization (both analytic and computational)

Example Gradient Calculation

Suppose

$$
\begin{aligned}
f(\boldsymbol{x}) &= f(x_1, x_2, \ldots, x_n) \\
&= x_1^2 + x_2^2 + \ldots + x_n^2 \\
&= \sum_{i=1}^{n} x_i^2
\end{aligned}
$$

Then $\nabla f(\boldsymbol{x}^*)$ is

$$
\nabla f(\boldsymbol{x}^*) = (2x_1^*, 2x_2^*, \ldots, 2x_n^*)
$$

So if $\boldsymbol{x}^* = (3, 3, \ldots, 3)$ then

$$
\begin{aligned}
\nabla f(\boldsymbol{x}^*) &= (2 * 3, 2 * 3, \ldots, 2 * 3) \\
&= (6, 6, \ldots, 6)
\end{aligned}
$$

Second Partial Derivative

Definition

Suppose $f : X \to \Re$ *where* $X \subset \Re^n$ *and suppose that* $\frac{\partial f(x_1, x_2, \ldots, x_n)}{\partial x_i}$ *exists. Then we define,*

$$\frac{\partial^2 f(\boldsymbol{x})}{\partial x_j \partial x_i} \equiv \frac{\partial}{\partial x_j}\left(\frac{\partial f(\boldsymbol{x})}{\partial x_i}\right)$$

- Second derivative could be with respect to x_i or with some other variable x_j
- Nagging question: does order matter?

Second Partial Derivative: Order Doesn't Matter

Theorem

Young's Theorem Let $f : X \to \Re$ with $X \subset \Re^n$ be a twice differentiable function on all of X. Then for any i, j, at all $\boldsymbol{x}^* \in X$,

$$\frac{\partial^2}{\partial x_i \partial x_j} f(\boldsymbol{x}^*) = \frac{\partial^2}{\partial x_j \partial x_i} f(\boldsymbol{x}^*)$$

Second Order Partial Derivates

$$f(\boldsymbol{x}) = x_1^2 x_2^2$$

Then,

$$\begin{aligned}
\frac{\partial^2}{\partial x_1 \partial x_1} f(\boldsymbol{x}) &= 2x_2^2 \\
\frac{\partial^2}{\partial x_1 \partial x_2} f(\boldsymbol{x}) &= 4x_1 x_2 \\
\frac{\partial^2}{\partial x_2 \partial x_2} f(\boldsymbol{x}) &= 2x_1^2
\end{aligned}$$

Hessians

Definition

Suppose $f : X \rightarrow \Re^1$, $X \subset \Re^n$, with f a twice differentiable function. We will define the Hessian matrix as the matrix of second derivatives at $\boldsymbol{x}^* \in X$,

$$\boldsymbol{H}(f)(\boldsymbol{x}^*) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1}(\boldsymbol{x}^*) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(\boldsymbol{x}^*) & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\boldsymbol{x}^*) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(\boldsymbol{x}^*) & \frac{\partial^2 f}{\partial x_2 \partial x_2}(\boldsymbol{x}^*) & \dots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(\boldsymbol{x}^*) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\boldsymbol{x}^*) & \frac{\partial^2 f}{\partial x_n \partial x_2}(\boldsymbol{x}^*) & \dots & \frac{\partial^2 f}{\partial x_n \partial x_n}(\boldsymbol{x}^*) \end{pmatrix}$$

- Hessians are symmetric
- They describe curvature of a function (think, how bended)
- Will be the basis for second derivative test for multivariate optimization

An Example

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Suppose $f : \Re^3 \to \Re$, with

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$$f(x_1, x_2, x_3) \quad = \quad x_1^2 x_2^2 x_3^2$$

$$\nabla f(\boldsymbol{x}) \quad = \quad (2x_1 x_2^2 x_3^2, 2x_1^2 x_2 x_3^2, 2x_1^2 x_2^2, x_3)$$

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Suppose $f : \Re^3 \to \Re$, with

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$$\boldsymbol{H}(f)(\boldsymbol{x}) \quad = \quad \begin{pmatrix} 2x_2^2 x_3^2 & 4x_1 x_2 x_3^2 & 4x_1 x_2^2 x_3 \\ 4x_1 x_2 x_3^2 & 2x_1^2 x_3^2 & 4x_1^2 x_2 x_3 \\ 4x_1 x_2^2 x_3 & 4x_1^2 x_2 x_3 & 2x_1^2 x_2^2 \end{pmatrix}$$

Functions with Multidimensional Codomains

Definition

Suppose $f : R^m \rightarrow R^n$. We will call f a multivariate function. We will commonly write,

$$f(\boldsymbol{x}) = \begin{pmatrix} f_1(\boldsymbol{x}) \\ f_2(\boldsymbol{x}) \\ \vdots \\ f_n(\boldsymbol{x}) \end{pmatrix}$$

Example Functions

Suppose $f : \Re \rightarrow \Re^2$,

$$f(t) \quad = \quad (t^2, \sqrt(t))$$

Example Functions

Suppose $f : \Re^2 \rightarrow \Re^2$ defined as

$$f(r, \theta) = \begin{pmatrix} r\cos\theta \\ r\sin\theta \end{pmatrix}$$

Example Functions

Suppose we have some policy $\boldsymbol{x} \in \Re^M$. Suppose we have N legislators where legislator i has utility

$$U_i(\boldsymbol{x}) \;\; = \;\; \sum_{j=1}^{M} -(x_j - \mu_{ij})^2$$

We can describe the utility of all legislators to the proposal as

$$f(\boldsymbol{x}) \;\; = \;\; \begin{pmatrix} \sum_{j=1}^{M} -(x_j - \mu_{1j})^2 \\ \sum_{j=1}^{M} -(x_j - \mu_{2j})^2 \\ \vdots \\ \sum_{j=1}^{M} -(x_j - \mu_{Nj})^2 \end{pmatrix}$$

Jacobian

Definition

Suppose $f : X \to \Re^n$, where $X \subset \Re^m$, with f a differentiable function. Define the Jacobian of f at $\boldsymbol{x}$ as

$$\boldsymbol{J}(f)(\boldsymbol{x}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_m} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_m} \end{pmatrix}$$

Remark: The Jacobian matrix is important because if the function f is differentiable at a point x, then the Jacobian matrix defines a linear map $R^n \to R^m$, which is the best (pointwise) linear approximation of the function f near the point x. This linear map is thus the generalization of the usual notion of derivative, and is called the derivative or the differential of f at x.

Example of Jacobian

$$f(r,\theta) = \begin{pmatrix} r\cos\theta \\ r\sin\theta \end{pmatrix}$$

$$\boldsymbol{J}(f)(r,\theta) = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}$$